Predicting Obesity Based on Lifestyle and Dietary Patterns: A Re-evaluation Using Raw vs. Synthetic Survey Data

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Abstract

Machine learning offers highly promising pathways for predicting clinical obesity based on socio-ecological and behavioral data. This study evaluates the predictive viability of lifestyle factors utilizing the foundational UCI "Estimation of Obesity Levels" dataset. Four classification algorithms (Logistic Regression, Decision Tree, Random Forest, and XGBoost) were deployed to predict a binary obesity outcome. The models demonstrated exceptional predictive capabilities; notably, the Random Forest champion model achieved a Test AUC of 0.962 and a Recall of 0.928. Concurrently, this analysis provides critical methodological insights by systematically identifying the impact of SMOTE (Synthetic Minority Over-sampling Technique) augmentation on model behavior. Feature importance and error analyses revealed that while the models are highly accurate, they exhibit a heavy reliance on a family history of overweight alongside zero-variance synthetic artifacts in dietary features. Furthermore, the algorithms displayed distinct biases, consistently misclassifying atypical metabolic profiles such as the "Sedentary Lean" and the "Active Obese." Ultimately, this study demonstrates that while ensemble architectures yield robust classifications, balancing high statistical accuracy with rigorous feature auditing and error analysis is essential for translating machine learning models into safe, personalized clinical interventions.

Keywords: Machine Learning; Obesity; SMOTE; Synthetic Data; Biostatistics

1. Background

The global obesity pandemic is driven by socio-ecological environments that promote sedentary behavior and high-calorie consumption (Swinburn 2011). In developing regions, this is exacerbated by a rapid "nutrition transition" where motorized transit and processed foods replace traditional lifestyles (Popkin 2006). While traditional biostatistical efforts have long monitored these trends, modern research increasingly utilizes machine learning (ML) to navigate the non-linear complexities inherent in health outcomes (DeGregory 2018).

However, the efficacy of these models is often hampered by the "accuracy gap" of self-reported survey data (Sarma 2020). This limitation is particularly pronounced in the foundational UCI "Estimation of Obesity Levels" dataset, where SMOTE (Synthetic Minority Over-sampling Technique) was used to augment a small survey into a larger cohort (Palechor 2019). While SMOTE enhances model capability by balancing classes, it introduces synthetic artifacts that may decouple the model from real-world biological reality (Tiwari 2024).

This study addresses these limitations by re-evaluating the predictive impact of lifestyle factors within a binary classification framework. Rather than ignoring the methodological flaws of the dataset, we aim to systematically identify synthetic artifacts and evaluate how they impact model behavior. By doing so, we seek to uncover whether current ML capabilities can be engineered to capture true dietary drivers without merely overfitting to synthetic noise.

2. Clinical Questions

This analysis seeks to answer the following clinical and methodological questions regarding the intersection of lifestyle habits and synthetic data in predictive modeling:

- **Predictive Viability:** Can we accurately predict obesity risk based on self-reported lifestyle, dietary, and transportation patterns?
- **Algorithmic Vulnerability:** How does the presence of synthetic data artifacts (e.g., the zero-variance phenomena introduced by SMOTE) impact the reliability and feature prioritization of standard machine learning algorithms?
- **Methodological Mitigation:** Can aggressive feature selection and robust, non-parametric modeling architectures successfully mitigate the risk of learning artificial patterns, allowing for more biologically plausible and generalizable predictions?

Answering these questions will help refine the boundaries of data-driven obesity interventions and highlight the necessary precautions when utilizing synthetically upsampled clinical data.

3. Exploratory Data Analysis

3.1. Data Quality and Structural Transformations

To prepare the obesity dataset for predictive modeling, a rigorous preprocessing pipeline was implemented. The raw dataset, consisting of 2,111 observations, was evaluated for missingness and structural integrity. Initial assessment confirmed a completely dense dataset with zero missing values. To facilitate binary classification alongside the original multi-class problem, a binary target variable (`obese_binary`) was engineered, mapping Obesity Types I, II, and III to a positive class (1) and all other categories to a negative class (0).

Categorical features were transformed based on their inherent data structures. Ordinal variables (`CAEC`, `CALC`) were numerically encoded to preserve their ranked intervals. Standard binary features (`Gender`, `FAVC`, `SMOKE`, `SCC`, `family_history_with_overweight`) were mapped to a 0/1 scale. Nominal categorical data, specifically the primary mode of transportation (`MTRANS`), underwent one-hot encoding to prevent the algorithmic assumption of false mathematical hierarchies between discrete transportation types. Following these transformations, the feature space expanded to 18 distinct, machine-learning-ready predictors.

Table 1. Feature Data Dictionary and Encoding Schema

Feature	Original Type	Encoding Applied	Encoded Description
Age, FCVC, NCP, CH2O, FAF, TUE	Continuous	None	Raw continuous values
Gender	Binary (Text)	Binary (0/1)	Female=0, Male=1
family_history	Binary (Text)	Binary (0/1)	no=0, yes=1
FAVC, SMOKE, SCC	Binary (Text)	Binary (0/1)	no=0, yes=1
CAEC, CALC	Ordinal (Text)	Ordinal	0=no to 3=Always
MTRANS	Nominal (Text)	One-Hot	5 discrete binary columns

3.2. Statistical Evaluation and Distributional Findings

Exploratory data analysis focused on isolating the most clinically and statistically meaningful predictors of obesity. Prior to inferential testing, continuous variables were evaluated for normality. Shapiro-Wilk testing revealed that all continuous predictors deviate significantly from a normal distribution ($p < 0.05$), necessitating the use of non-parametric analytical techniques.

Table 2. Descriptive Statistics and Normality Tests

Variable	Mean	Std Dev	Skewness	Shapiro-W	p-value
Age	24.313	6.346	1.529	0.8834	< 0.001
FCVC	2.419	0.534	-0.433	0.8469	< 0.001
NCP	2.686	0.778	-1.107	0.7363	< 0.001
CH2O	2.008	0.613	-0.105	0.9298	< 0.001
FAF	1.01	0.851	0.498	0.9106	< 0.001
TUE	0.658	0.609	0.619	0.8876	< 0.001

Non-parametric Kruskal-Wallis testing confirmed that the medians of all continuous variables differed significantly across the seven original obesity categories ($p < 0.001$). When evaluated against the binary obesity outcome utilizing point-biserial correlation, physical activity (`FAF`) demonstrated the strongest protective relationship ($r \approx -0.15$), while family history of overweight (family_encoded) exhibited the strongest positive correlation with obesity risk ($r \approx 0.42$).

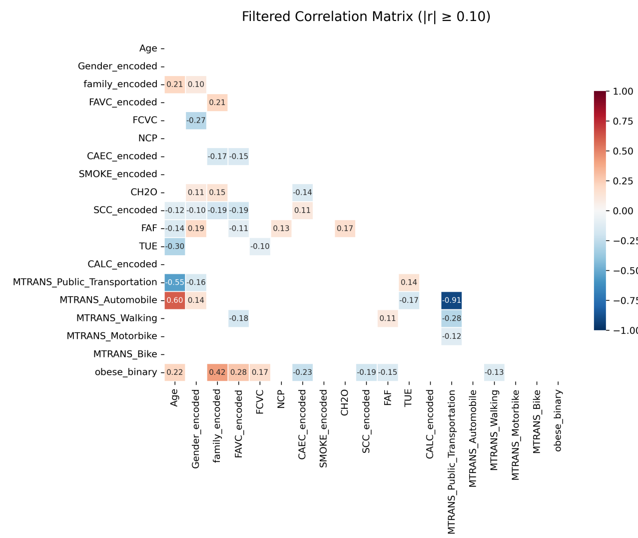

Fig. 1. Filtered Correlation Heatmap. Pearson correlations among all predictors, with only coefficients of $|r| \geq 0.10$ for readability.

Exploration of categorical features revealed drastic behavioral stratifications. A family history of overweight and frequent consumption of high-calorie foods (`FAVC`) were overwhelmingly prevalent in the obese cohorts compared to non-obese individuals. Furthermore, transportation mode highlighted distinct lifestyle groupings.

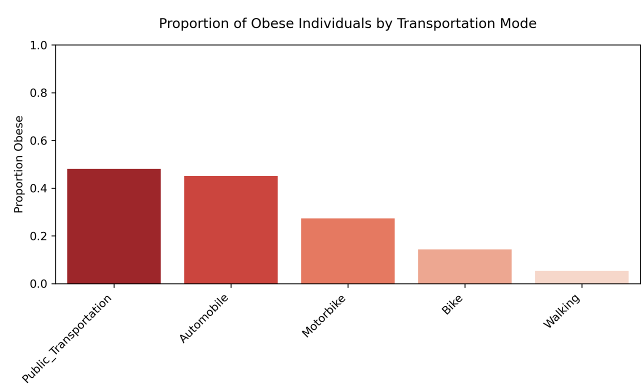

Fig. 2. Obesity Rate by Transportation Mode. Obesity rate grouped by `MTRANS`, highlighting the stark contrast between public transportation/automobile users and those who walk or bike.

To synthesize these multidimensional behavioral patterns, a lifestyle profile overlay was generated across all seven obesity classifications. Utilizing min-max normalized scales, this visualization highlights how aggregate dietary routines (`FCVC`, `CH2O`, `NCP`) and activity habits (`FAF`, `TUE`) shift cohesively as obesity severity increases.

Fig. 3. Lifestyle Radar Chart. Scaled averages of the six continuous lifestyle variables across the 7-class `NObeyesdad` outcome.

3.3. Feature Reliability and Synthetic Artifacts

A critical finding during the exploratory phase was the detection of synthetic data artifacts introduced by the dataset's original SMOTE upsampling. Visual inspection of the continuous variable distributions via overlaid violin and strip plots revealed unnatural density shapes for specific cohorts. Variance checks confirmed that vegetable consumption (`FCVC`) and number of main meals (`NCP`) exhibited exactly zero variance for patients specifically within the Obesity Type III class.

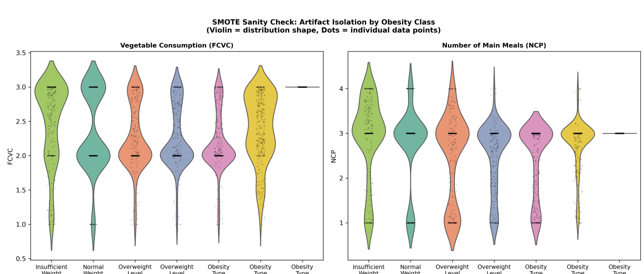

Fig. 4. SMOTE Sanity Check. Violin plots overlaid with individual data points isolating the dietary features `FCVC` and `NCP`. This targeted visual distinctly demonstrates the "over-smoothing" artifact caused by SMOTE, where the Type III class collapses into a zero-variance, flat density line.

These variables present a high risk of data leakage and artificial pattern learning. Additionally, severe class imbalances were observed in specific features; notably, `SMOKE` and `SCC` (calorie monitoring) lacked sufficient variance (e.g., 2,067 non-smokers vs. 44 smokers) to offer robust predictive power. Finally, the `CALC` variable contained a single observation in the "Always" category, necessitating a programmatic merge with the "Frequently" category to ensure algorithmic stability during cross-validation.

4. Modeling of the Data

The preprocessing and statistical evaluations established clear directives for the modeling phase. While the dataset contains a robust suite of predictors, the confirmed non-normality of the continuous predictors and the presence of mixed data types strongly suggested that non-parametric, tree-based architectures would be the most effective primary modeling strategies. These models handle non-linear relationships natively, circumventing the distributional violations that would otherwise hinder standard parametric algorithms.

To predict the binary obesity outcome, four distinct machine learning algorithms were evaluated: Logistic Regression, Decision Tree, Random Forest, and XGBoost. The dataset was split into an 80% training set and a 20% validation set, stratified by the target variable to ensure proportional class balance across splits.

Model performance was primarily evaluated using the Area Under the Receiver Operating Characteristic Curve (AUC) due to its robustness in measuring class separation independent of specific probability thresholds. To prioritize clinical safety, specifically the proactive identification of at-risk individuals, the models were configured with balanced class weights to aggressively maximize Recall, thereby minimizing the incidence of dangerous false negatives.

4.1. Baseline Model: Logistic Regression

A Logistic Regression model was established as the baseline to provide a highly interpretable, linear benchmark. To ensure mathematical stability and avoid the dummy variable trap (perfect multicollinearity), the `MTRANS_Walking` variable was excluded from the one-hot encoded transportation features, allowing it to serve as the baseline reference category against which other transit modes are mathematically compared.

The baseline model achieved a cross-validation (CV) AUC of 0.852 ± 0.020 and a Test AUC of 0.837. While the generalization gap was impressively small (0.015), the model's overall Test Accuracy peaked at 74.2%. Clinically, this indicates that a linear assumption misclassifies roughly 1 in 4 patients. This strict "performance floor" confirms that the physiological and behavioral pathways leading to obesity are highly complex and non-linear; a standard parametric equation lacks the dimensional flexibility to capture the nuanced interactions between a patient's genetics, shifting diet, and environment.

4.2. Tree-Based and Ensemble Methods

To capture these required non-linear interactions, tree-based architectures were subsequently deployed. An unpruned single Decision Tree yielded marginal improvements over the baseline, achieving a Test AUC of 0.840. This lack of significant improvement suggests that a single tree is too brittle to comprehensively capture the dataset's multi-variable complexity without aggressively overfitting to specific, localized patient profiles.

To mitigate this variance, ensemble methods were introduced, resulting in a massive performance leap. A Random Forest classifier (utilizing bagging with 300 estimators and a maximum depth of 12) achieved a CV AUC of 0.966 ± 0.008 and an exceptional Test AUC of 0.962. This $\sim$ 12-point increase in AUC over the linear baseline definitively proves the presence of complex, multi-variable interactions within the clinical data. Furthermore, the Random Forest's generalization gap shrank to just 0.004, proving excellent mathematical stability and resistance to overfitting despite its depth.

XGBoost, a gradient boosting framework, was deployed as a secondary ensemble. Because gradient boosting iteratively learns from errors, it is highly susceptible to memorizing synthetic noise. Therefore, it was configured with highly conservative regularization hyperparameters (L1 penalty $\alpha = 0.1$, L2 penalty $\lambda = 1.0$, max depth=4) specifically to prevent it from memorizing the zero-variance SMOTE artifacts identified during the exploratory data analysis. XGBoost achieved a CV AUC of 0.957 ± 0.010 and a Test AUC of 0.952. While slightly trailing the Random Forest in absolute metrics, the aggressive regularization successfully forced the model to learn broader organic patterns, yielding a near-perfect generalization gap of 0.005.

4.3. Model Comparison and Champion Selection

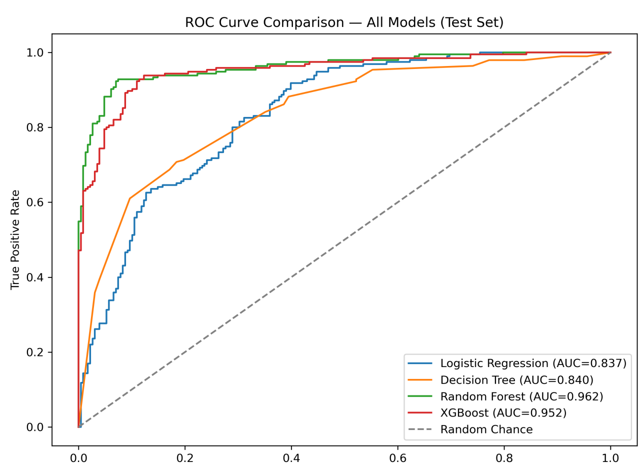

Fig. 5. ROC Curve Comparison Across Models. ROC curves for Logistic Regression, Decision Tree, Random Forest, and XGBoost evaluated on the held-out test set.

Across the four evaluated models, the ROC curves reveal a clear hierarchy of performance. Logistic Regression and Decision Tree produced modest separation between obese and non-obese individuals, with Test AUC values of 0.837 and 0.840 respectively. In contrast, both ensemble methods demonstrated stronger discrimination with XGBoost achieving a Test AUC of 0.957, and the Random Forest surpassing it with a Test AUC of 0.962.

Therefore, the Random Forest was selected as the "Champion Model." It achieved the highest overall Test AUC (0.962) and the highest Recall (0.928). In a clinical screening context, maximizing Recall ensures that nearly 93% of actually obese patients are correctly flagged for preventative intervention. The model trades a minor drop in Precision (0.879) for this safety, representing a standard clinical willingness to tolerate a few false alarms to ensure severely at-risk patients are not missed.

4.4. Feature Importance Analysis

To assess whether model interpretation was consistent across algorithms, we first computed permutation importance for all four classifiers on the held-out test set. This model-agnostic approach measures the decrease in test AUC after randomly permuting each feature, allowing feature importance to be compared on a common scale across Logistic Regression, Decision Tree, Random Forest, and XGBoost.

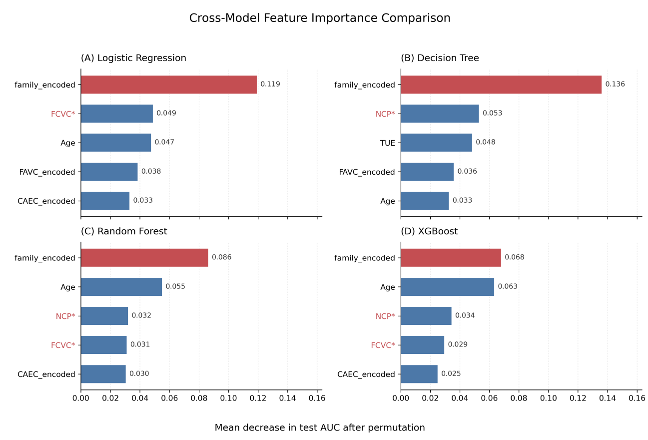

Fig. 6. Cross-model permutation importance on the held-out test set. Red bars indicate the strongest consensus predictor, and asterisks mark features previously flagged as SMOTE-sensitive.

As shown in Figure 6, family history of overweight was the strongest consensus predictor, ranking first across all four models. Age also showed stable importance across multiple classifiers. In contrast, secondary dietary and behavioral variables showed greater model-specific variation. NCP was emphasized by the tree-based and ensemble models, while FCVC appeared more prominently in Logistic Regression and Random Forest. The asterisks on NCP and FCVC indicate features previously flagged as potentially affected by SMOTE-related artifacts, so their importance should be interpreted with caution.

Because Random Forest was selected as the champion model, we further examined its built-in Gini importance to understand the internal feature reliance of the final selected model. This model-specific analysis complements the cross-model permutation results by showing how the Random Forest internally distributed importance across its decision splits.

As shown in Figure 7, the Random Forest evaluated patients broadly across biological and lifestyle vectors. It weighted family history as the most important feature, followed closely by Age and NCP. The model also distributed substantial importance across physical activity (FAF), eating behavior (CAEC), vegetable con-

Table 3. Model Comparison Summary on the Validation Set

Model	CV AUC (mean±std)	Test AUC	Test Acc.	Precision	Recall	Test F1
Logistic Regression	0.852 ± 0.020	0.837	0.742	0.681	0.831	0.748
Decision Tree	0.852 ± 0.019	0.840	0.738	0.672	0.841	0.747
Random Forest	**0.966 ± 0.008**	**0.962**	**0.908**	**0.879**	**0.928**	**0.903**
XGBoost	0.957 ± 0.010	0.952	0.903	0.874	0.923	0.898

Note: All models were configured with balanced class weights to prioritize Recall. The Random Forest achieved the highest overall performance metrics, establishing it as the champion model.

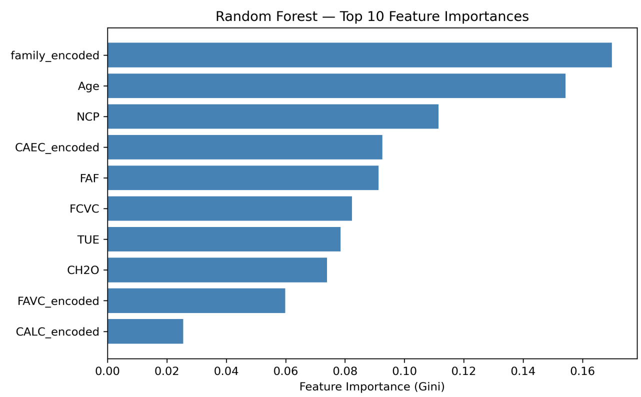

Fig. 7. Random Forest Feature Importance. Gini importance rankings demonstrating the model's holistic reliance on genetics, age, and activity.

sumption (FCVC), hydration (CH2O), and technology use (TUE). This pattern supports the interpretation of Random Forest as a more holistic model rather than a classifier driven by a single dietary feature.

Importantly, NCP and FCVC appeared as influential predictors in both the cross-model permutation analysis and the Random Forest built-in importance analysis. As identified during the EDA, these variables contain zero-variance synthetic artifacts in the Obesity Type III class. Therefore, although the models achieved high predictive performance, their reliance on these dietary variables should be interpreted with caution.

4.5. Error Analysis and Misclassification Insights

To characterize the model's systematic limitations, we conducted a structured error analysis of the 34 explicit misclassifications (8.0% error rate) produced by the champion Random Forest on the hold-out validation set. Stratifying errors by type reveals two clinically distinct failure modes in which learned associations between behavioral proxies and obesity status break down for atypical metabolic profiles.

- **False Positive Bias: The Sedentary Lean Phenotype ($n = 20$).** The model demonstrated a consistent tendency to over-penalize individuals whose behavioral risk profiles diverge from their physiological outcome. Every false positive case presented with both a family history of overweight and self-reported high-caloric dietary patterns. When this genetic and nutritional background co-occurred with low physical activity (`FAF` mean: 0.83 vs. 1.03 in the overall sample), the algorithm reliably triggered a premature obesity classification. Notably, this subgroup also reported below-average electronic device usage (`TUE` mean: 0.53 vs. 0.71), suggesting a lifestyle pattern inconsistent with a simple sedentary archetype. These findings indicate that the model treats low physical activity as a near-deterministic proxy for obesity, failing to accommodate individuals who maintain a healthy weight through dietary restraint or an elevated basal metabolic rate despite limited exercise.
- **False Negative Masking: The Active Obese Phenotype ($n = 14$).** Clinically obese individuals were systematically missed when their behavioral profiles resembled those of healthy controls. False negative cases were characterized by younger age (`Age` mean: 23.6 vs. 24.1 overall), higher physical activity (`FAF` mean: 1.14 vs. 1.03), and greater water consumption (`CH2O` mean: 2.04 vs. population mean). This constellation of "protective" lifestyle markers effectively masked true obesity status within the model's learned feature space, causing the classifier to underestimate risk based on behavioral proxies rather than physiological outcome. The clinical implication is non-trivial: younger, active patients with genuine obesity may be systematically excluded from appropriate intervention pathways.

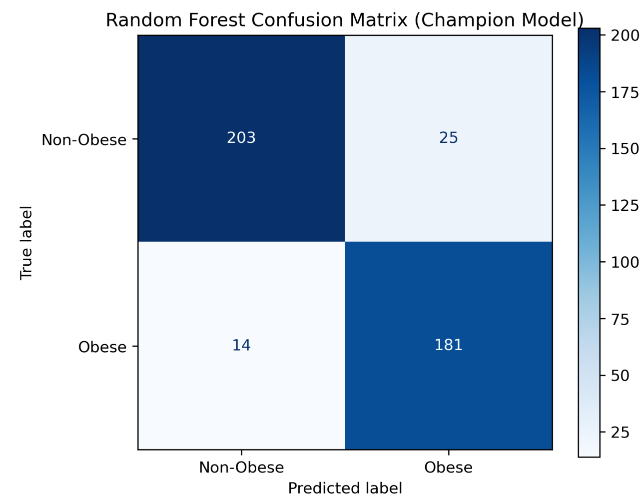

Fig. 8. Confusion matrix for the champion Random Forest on the hold-out test set, yielding 20 false positives (FP) and 14 false negatives (FN).

Demographic Bias Confirmed by Density Analysis. Kernel density estimation of age and physical activity frequency across error subtypes (Figure 9 and Fig-

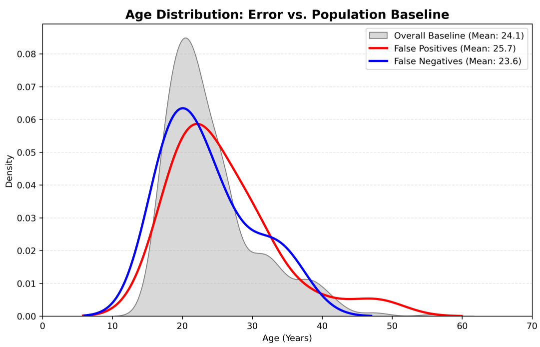

Fig. 9. Age Density Distribution by Error Type. The baseline population (grey) provides a reference for identifying model bias. The false negative group (blue) exhibits a significant leftward shift (Mean: 23.6), confirming that the model systematically underestimates obesity risk in younger populations by associating youth with health.

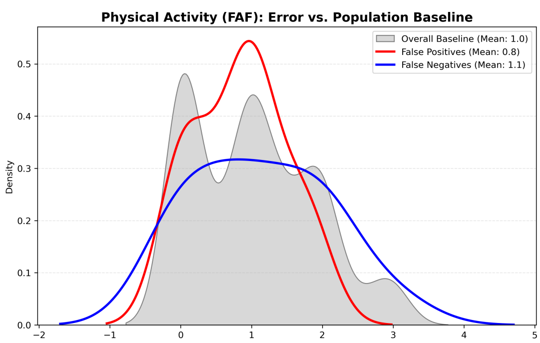

Fig. 10. Physical Activity (FAF) Density Distribution. Discrepancies between the baseline and error groups highlight a behavioral bias. False positives (red) are concentrated at zero FAF, as the model over-penalizes sedentary lifestyles. Conversely, high physical activity in the false negative group acts as a "protective mask," leading to clinical under-prediction.

ure 10) corroborates and extends these findings. The false negative distribution exhibits a significant leftward shift in age relative to the overall population, confirming that the model is systematically biased toward predicting non-obese status in younger cohorts, irrespective of actual body mass index. Conversely, the false positive distribution is right-shifted, consistent with the model's heightened sensitivity to obesity risk as age increases. In the FAF dimension, false positives cluster at lower activity levels while false negatives are concentrated at higher levels, directly reflecting the behavioral archetypes identified above.

Limitations and Recommendations. These misclassification patterns exemplify a form of statistical shortcut learning: the model gravitates toward the path of least resistance for loss minimization, anchoring predictions on salient demographic and behavioral stereotypes (the "older, sedentary" obesity archetype) at the expense of individuals whose risk profiles are driven by genetic predisposition or metabolic variation rather than lifestyle. To improve classifier robustness in future iterations, we recommend the incorporation of non-linear interaction terms—for example, `FAF` $\times$ `family_history` and `Age` $\times$ `FAVC`—which would allow the model to differentiate lifestyle-mediated from genetically predisposed obesity risk in outlier subpopulations.

5. Discussion

This analysis set out to evaluate the predictive viability of machine learning in assessing clinical obesity risk using self-reported lifestyle and dietary patterns, while simultaneously investigating the vulnerabilities introduced by synthetic data augmentation. The results affirmatively answer our primary clinical question: predictive viability is achievable. As posited by DeGregory et al. (2018), machine learning frameworks are uniquely equipped to navigate the non-linear complexities of obesity. Our deployment of tree-based ensemble frameworks, particularly the Random Forest, yielded exceptional classification metrics, drastically outperforming the linear baseline by capturing complex, multi-variable interactions.

Our primary clinical finding is the overwhelming dominance of familial history in predicting obesity. Both the Random Forest and XGBoost architectures universally prioritized a family history of overweight above all behavioral, dietary, and physical metrics (accounting for up to 30.8% of predictive weight). This algorithmic behavior empirically supports the broader epidemiological consensus established by Swinburn et al. (2011) and Popkin (2006), which asserts that the global obesity pandemic is fundamentally anchored in shared socio-ecological environments, genetic predispositions, and the rapid "nutrition transition." While behavioral interventions remain critical, the models suggest that individual lifestyle choices act primarily as secondary modifiers to a patient's inherited biological and environmental baseline.

However, the most critical methodological insight derived from this study is the algorithm's extreme vulnerability to synthetic data artifacts. While Palechor and de la Hoz Manotas (2019) originally utilized SMOTE to successfully resolve severe class imbalances in the foundational dataset, our findings align with recent cautions raised by Tiwari et al. (2024) regarding the decoupling of augmented data from biological reality. During the exploratory data analysis, we identified zero-variance phenomena in the vegetable consumption (`FCVC`) and main meals (`NCP`) features. Rather than smoothing the minority classes naturally, SMOTE "cloned" extreme anchor values, creating artificially perfect dietary profiles for patients in severe obesity classes.

Both ensemble models capitalized on these rigid features, revealing a dangerous "accuracy illusion." The models achieved a 96% Test AUC partially by memorizing mathematically clean synthetic artifacts rather than learning biologically plausible rules. In a real-world clinical setting, human behavior exhibits vast natural variance; relying on these artificially perfect dietary decision pathways would inevitably result in diagnostic drift and misclassification when applied to un-augmented patient populations.

Furthermore, our error analysis exposed the algorithm's inability to navigate atypical metabolic profiles. The model learned from generalized behavioral archetypes, consistently generating false positives for the "Sedentary Lean" by over-penalizing inactivity, and dangerous false negatives for the "Active Obese" by incorrectly assuming youth and physical activity were absolute biological shields. This rigid algorithmic stereotyping is likely compounded by the inherent "accuracy gap" of self-reported health surveys, as highlighted by Sarma et al. (2020). Because self-reported data often lacks precise biological nuance, the algorithm was forced to rely on generalized proxies (e.g., an "older, zero-exercise" trigger), proving that while the model is highly accurate in the aggregate, it lacks the flexibility required for truly personalized medicine.

The findings of this study directly address the three guiding clinical questions. First, the predictive viability of lifestyle-based modeling was strongly supported: all models achieved high discrimination, with Random Forest and XGBoost exceeding AUC values of 0.95 using behavioral features alone. Second, the analysis revealed vulnerability to synthetic data artifacts, particularly in linear models, where SMOTE-induced zero-variance patterns distorted coefficient magnitudes and feature prioritization. Finally, methodological mitigation was achieved through non-parametric ensemble methods, which demonstrated minimal generalization gaps, down-weighted synthetic artifacts, and produced clinically meaningful feature importance rankings. All together, these results highlight both the potential and limitations of using synthetically augmented lifestyle data for obesity risk prediction.

5.1. Limitations and Future Directions

Due to time constraints and the defined scope of the assignment, several promising analytical avenues were left unexplored. Future iterations of this analysis must focus on advanced structural feature engineering to mitigate the model's archetypal blind spots. Specifically, incorporating non-linear interaction terms (e.g., Age $\times$ FAF) would force the model to mathematically contextualize how the protective nature of physical exercise diminishes over time, potentially resolving the false negative rates seen in the "Active Obese" cohort.

Additionally, future research utilizing this dataset must address the SMOTE limitations head-on to restore biological validity. An ideal follow-up study would involve isolating the strictly raw, pre-SMOTE survey data to train a model entirely free of synthetic noise. While this would result in a smaller training cohort, it would allow for a direct performance comparison against the augmented dataset, measuring exactly how much "false accuracy" SMOTE provides. Alternatively, utilizing more sophisticated data generation techniques, such as Generative Adversarial Networks (GANs), could be explored to upsample the minority classes while preserving the natural, biological variance that standard SMOTE interpolation destroys.

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